

## Assignment 5: Differential Forms

to be handed in online by Sunday, June 28th

Students taking the “[...] Black Holes” physics course are only expected to submit the exercises marked with an asterisk\*, but are obviously invited to try all of them. Students taking the “[...] Classical Systems” maths course are expected to submit (almost) all exercises.

If you skip an exercise, be sure to read it at least, since you might need its result later.

We have hidden most hints as white text, to encourage you to try to solve the problems without looking at them. If you are stuck, simply copy the white text somewhere else to reveal it.

Feel free to use **Mathematica**, but submit the notebook or screenshots of it.

### Exercise 1: The Wedge Product

Let  $\omega$  be a  $p$ -form,  $\eta$  a  $q$ -form and  $\theta$  an  $r$ -form on some manifold  $M$ . Show that the wedge product satisfies the following properties

a)  $\omega \wedge \eta = (-1)^{p \cdot q} \eta \wedge \omega$ .

b)  $(\omega \wedge \eta) \wedge \theta = \omega \wedge (\eta \wedge \theta)$ .

### Exercise 2: The Exterior Derivative

Let  $\omega$  be a  $p$ -form and  $\eta$  a  $q$ -form on some manifold  $M$ . Show that the exterior derivative satisfies the following properties

a)  $d(d\omega) = 0$ .

b)  $d(\omega \wedge \eta) = (d\omega) \wedge \eta + (-1)^p \omega \wedge (d\eta)$ .

c)  $d(\varphi^* \omega) = \varphi^*(d\omega)$ , where  $\varphi : M \rightarrow N$  is a map between manifolds  $M$  and  $N$ , and  $\varphi^*$  is its pullback.

### Exercise 3: The Hodge Star Operator

Let  $\omega$  and  $\eta$  be  $p$ -forms. Recall that we defined the Hodge star operator as

$$(\star \omega)_{\mu_1 \dots \mu_{D-p}} = \frac{1}{p!} \varepsilon^{\nu_1 \dots \nu_p}_{\mu_1 \dots \mu_{D-p}} \omega_{\nu_1 \dots \nu_p}, \quad (1)$$

where  $\varepsilon$  is the Levi-Civita *tensor*,

$$\varepsilon_{\mu_1 \dots \mu_D} = \sqrt{-\det g} \, \epsilon_{\mu_1 \dots \mu_D}, \quad \varepsilon^{\mu_1 \dots \mu_D} = -\frac{1}{\sqrt{-\det g}} \epsilon^{\mu_1 \dots \mu_D}, \quad (2)$$

with  $\epsilon$  being the usual Levi-Civita symbol. Notice the extra minus sign in the right expression, caused by having raised the indices on  $\epsilon$  with a Lorentzian metric.

We now define

$$\langle \omega, \eta \rangle := \int_{\mathcal{M}} \omega \wedge \star \eta. \quad (3)$$

a) Prove (*extra exercise*)

$$\varepsilon_{\alpha_1 \dots \alpha_p \gamma_1 \dots \gamma_{D-p}} \varepsilon^{\beta_1 \dots \beta_p \gamma_1 \dots \gamma_{D-p}} = -p!(D-p)! \delta_{[\alpha_1}^{\beta_1} \dots \delta_{\alpha_p]}^{\beta_p} \quad (4)$$

b) Prove

$$\langle \omega, \eta \rangle = \frac{1}{p!} \int d^D x \sqrt{-g} \omega_{\alpha_1 \dots \alpha_p} \eta^{\alpha_1 \dots \alpha_p}. \quad (5)$$

c) Prove

$$\langle \omega, \eta \rangle = \langle \eta, \omega \rangle. \quad (6)$$

d)\* Prove that for a zero form  $\phi$

$$\langle d\phi, d\phi \rangle = \int_{\mathcal{M}} d^D x \sqrt{-g} \partial^\mu \phi \partial_\mu \phi. \quad (7)$$

#### Exercise 4: Stokes' Theorem

Given a  $p-1$  form  $\omega$  and a  $p$ -dimensional submanifold  $\Sigma$  with boundary  $\partial\Sigma$  we have that

$$\int_{\Sigma} d\omega = \oint_{\partial\Sigma} \omega. \quad (8)$$

a)\* Show that, in three space dimensions eq. (8) is equivalent to Stokes' theorem

$$\int_{\Sigma} d^2 x \vec{n} \cdot (\vec{\nabla} \times \vec{A}) = \oint_{\partial\Sigma} d\vec{x} \cdot \vec{A}, \quad (9)$$

by taking  $\omega = A_\mu dx^\mu$ .

b)\* Similarly, again in three dimensions, show that taking  $\omega = \star(A_\mu dx^\mu)$  and integrating over a three-dimensional region  $U$  instead of  $\Sigma$  one obtains Gauss' law

$$\int_U d^3 x \vec{\nabla} \cdot \vec{A} = \oint_{\partial U} d^2 x \vec{n} \cdot \vec{A} \quad (10)$$

#### Exercise 5: $p$ -form Calculus

Solve the following problems and simplify the results as much as possible.

a)\* Compute and simplify

$$(dx + dy) \wedge (f(y) dx + g(y) dy) \quad (11)$$

$$d(xy dz) \quad (12)$$

$$(x dy + y dz + z dx) \wedge d(x dy + y dz + z dx) \quad (13)$$

**b)\*** Which of the following forms  $\omega$  is closed, i.e.  $d\omega = 0$ ? Which of them is exact, i.e.  $\omega = d\eta$  for some  $\eta$ ?

$$y dx + x dy \quad (14)$$

$$(x^2 + y^2) dx \quad (15)$$

$$\frac{y dx - x dy}{x^2 + y^2} \quad \text{on } \mathbb{R}^2 \setminus \{0\} \quad (16)$$

### Exercise 6: *M*-Theory

*M*-theory is a proposed quantum theory of gravity in  $D = 11$  spacetime dimensions. It arises from the strong coupling limit of string theory. At low energies, it is described by (the unique theory of) 11-dimensional supergravity, whose bosonic fields are the metric and a 4-form  $G = dC$ , where  $C$  is a 3-form potential. The action governing these fields is

$$S_M = \frac{1}{2} M_{\text{pl}}^9 \left[ \int d^{11}x \sqrt{-g} \left( R - \frac{1}{48} G_{\mu\nu\rho\sigma} G^{\mu\nu\rho\sigma} \right) - \frac{1}{6} \int C \wedge G \wedge G \right], \quad (17)$$

where  $M_{\text{pl}}$  denotes the Planck mass.

**a)\*** Show that, up to surface terms, this action is gauge invariant under  $C \rightarrow C + d\Lambda$ , where  $\Lambda$  is a 2-form.

**b)\*** Vary the metric to determine the Einstein equations for this theory.

**c)\*** Vary  $C$  to obtain the equation of motion for the 4-form,

$$d \star G = \frac{1}{2} G \wedge G. \quad (18)$$

*Hint:*